11. $\underbrace{\begin{bmatrix}1\\-1\\4\end{bmatrix}}_{v_1}, \underbrace{\begin{bmatrix}3\\-5\\7\end{bmatrix}}_{v_2}, \underbrace{\begin{bmatrix}-1\\5\\h\end{bmatrix}}_{v_3}$ $\boxed{\text{Solution:}}$ By defn, $\exists\, c_1, c_2, c_3 \in \mathbb{R}$ s.t. not all of c_1, c_2 and c_3 are 0 together and $c_1v_1 + c_2v_2 + c_3v_3 = 0$.

$$\therefore c_1\begin{bmatrix}1\\-1\\4\end{bmatrix} + c_2\begin{bmatrix}3\\-5\\7\end{bmatrix} + c_3\begin{bmatrix}-1\\5\\h\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

. By defn. of matrix equation $Ax=b$, we have:

$$\underbrace{\begin{bmatrix}1&3&-1\\-1&-5&5\\4&7&h\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}c_1\\c_2\\c_3\end{bmatrix}}_{x} = \underbrace{\begin{bmatrix}0\\0\\0\end{bmatrix}}_{b}$$

. By Theorem 3, this matrix equation has the same solution set as the system of equations corresponding to the augmented matrix $[A\ b]$

$$\begin{bmatrix}1&3&-1&0\\-1&-5&5&0\\4&7&h&0\end{bmatrix} \xrightarrow[+\text{Row}1]{\text{Row}2=\text{Row}2} \begin{bmatrix}1&3&-1&0\\0&-2&4&0\\4&7&h&0\end{bmatrix} \xrightarrow[-4\times\text{Row}1]{\text{Row}3=\text{Row}3} \begin{bmatrix}1&3&-1&0\\0&-2&4&0\\0&-5&h+4&0\end{bmatrix}$$

$$\xrightarrow[\times(-1/2)]{\text{Row}2=\text{Row}2} \begin{bmatrix}1&3&-1&0\\0&1&-2&0\\0&-5&h+4&0\end{bmatrix} \xrightarrow[+5\times\text{Row}2]{\text{Row}3=\text{Row}3} \begin{bmatrix}1&3&-1&0\\0&1&-2&0\\0&0&h-6&0\end{bmatrix}$$

Let, $h-6=0$. $\therefore h=6$. $\therefore$ The 3rd row is zero.

<u>Verification of whether the final matrix is in echelon form</u>

1. Row 1 and Row 2 are non-zero rows. Row 3 is a zero row. $\therefore$ The condition that all non-zero rows must be above the zero rows is True.

2. Row 1 Leading element $= 1$ in Column 1
   Row 2 Leading element $= 1$ in Column 2
   $\therefore$ The condition that for each row, the leading element must be in a column to the right of the columns of the leading elements of all rows above it is True.

3. Row 2 Column 1 $=$ Row 3 Column 1 $= 0$
   Row 3 Column 2 $= 0$
   $\therefore$ The condition that for each column containing a leading element, all positions below that position must contain 0 is True.

$\therefore$ The final matrix is in echelon form. $\square$